PIFA: Adversarially Trained, Forgery-Resistant Attribution of Virtualization-Obfuscated APT Malware

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Abstract—Attributing malware to the advanced persistent threat (APT) group that produced it must stay reliable after commercial obfuscation has rewritten the code, and against an adversary who forges a file's metadata to evade or misdirect attribution. However, static attributors of virtualization-obfuscated malware have been evaluated on splits that leak executables or builds between training and testing, and they read header or byte features that an adversary can rewrite, so no method attributes protected malware robustly. Therefore, we propose PIFA (Perturbation-Invariant, Forgery-resistant Attribution), a gradient-boosted attributor that reads every portable-executable header field but is trained on forged, copied, and edited copies of each executable, so it uses a field only where it agrees with the non-forgeable loader-relevant structure. We further develop ObfusFormer, a multi-view Transformer trained on each executable paired with its protected version, and study both on a leakage-free, build-grouped, adversary-aware benchmark of nine APT groups rebuilt from our earlier work. On unseen builds, PIFA attributed VMProtect-protected executables with a macro-F_1 of 91.2%, and stayed at 91.1%, 90.9%, and 91.0% under forgery, false-flag, and editing attacks. Clean-trained trees on all header fields were more accurate on untouched samples (92.5%) but a false flag cut them to 50.2%; trees on the loader-relevant fields alone stayed robust (90.9%) but fell to 85.3% under editing. PIFA is thus the most accurate attribution that survives every attack, above ObfusFormer (87.5%) and every image and byte network (at most 64.1%).

Index Terms—Adversarial training, code obfuscation, gradient boosting, malware attribution, portable executable, robustness, virtualization obfuscation.

1 Introduction

Machine learning is widely studied for malware detection and classification [1]. Static classifiers decide from the bytes of an executable without running it. They face two decisions, namely whether a file is malicious and which threat group produced it. Attributing advanced persistent threat (APT) malware to its group addresses the second decision [2]–[4] and guides how defenders respond to an intrusion. Commercial virtualization obfuscators such as VMProtect and Themida [5] complicate it. They translate selected code into bytecode for a virtual machine embedded in the protected file [6], rewriting the byte statistics that a static classifier learns from. An adversary can also rewrite the descriptive metadata of a file, or copy it from another group, so that a classifier relying on such metadata misattributes the sample. A robust attributor must therefore read features that survive obfuscation and that an adversary cannot forge. In the APT executables studied here, VMProtect raised the median file size from 180 KiB to 9,648 KiB and the mean byte entropy from 4.78 to 6.93 bits per byte, so accuracy on plain executables does not show whether a classifier can attribute protected ones.

Previous studies have rendered malware as images and classified the images with texture features [7], convolutional neural networks (CNNs) [8], attention modules [9], and Vision Transformers [10], [11]. For packed executables, Aghakhani *et al.* [12] showed that static classifiers did not generalize to packers absent from their training data. In our previous work [13], we protected APT malware with VMProtect and classified images of the protected files with Mal-CNN, a depth-wise separable CNN with spatial attention, and reported an accuracy of 93.55%. That work, however, left three problems open. First, in its released image splits, 820 of 1,015 test executables also contribute augmented images to the training set, and many executables come from one build, so a classifier can recognize an individual executable instead of attributing its group; attribution of builds made after the training builds was never tested. Second, it is unknown which parts of a protected file still identify its family, and which of them an

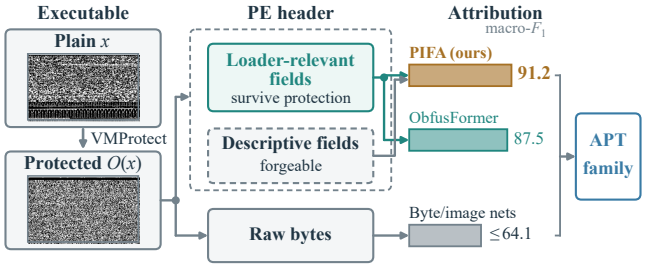


Fig. 1. Task and recommended attributor. PIFA reads the full PE header of a VMProtect-protected executable but is trained on forged, copied, and edited copies, so it attributes the APT family accurately (91.2%) and stays robust when the header is attacked, above the multi-view Transformer (ObfusFormer) and the image and byte networks on the raw bytes (protected-test macro-F_1, folds grouped by build). Clean-trained all-header trees score higher on untouched samples (92.5%) but collapse under a false flag (Table 6).

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adversary can rewrite, for example by editing the time stamp or copying the metadata of another group. Third, a model learns from one protection state at a time, and one trained on plain executables attributes protected executables poorly (Section 5.3).

Motivated by this, we propose PIFA (Perturbation-Invariant, Forgery-resistant Attribution), a gradient-boosted attributor that reads every portable executable (PE) header field but is trained to resist the header manipulations an adversary can apply. We split the PE header into the loader-relevant fields that Windows uses to run a program and the descriptive fields it ignores; VMProtect preserves the loader-relevant fields and an adversary cannot copy them from another group without changing how the program runs, whereas the descriptive fields carry a family signal but are forgeable. Rather than discard the descriptive fields, PIFA trains on forged, copied, and edited copies of every executable, so that it uses a descriptive field only where the field agrees with the non-forgeable loader-relevant structure, keeping the accuracy of the full header while staying robust under attack. To test whether a learned representation of the raw file can match it, we develop the Obfuscation-Paired Multi-View Transformer (ObfusFormer), which reads the loader-relevant fields, the byte distribution, and two 64-KiB windows, and which we train on each executable together with its protected version, treating the pair as two views of one sample in the manner of contrastive learning [14]. To measure both without the leakage and the forgery that earlier evaluations missed, we rebuild our corpus into a benchmark whose folds keep all executables of one build together, with a split by time and with test samples whose header fields are forged, copied from another group, or edited. We validate detection on 46,245 images derived from the EMBER benchmark [15] and attribution on 2,352 APT executables of nine groups with 2,187 protected counterparts.

Our contributions are summarized as follows:

- We propose PIFA, a gradient-boosted attributor that reads the full PE header but is trained on adversarial header manipulations to resist them. On unseen builds of nine APT groups, PIFA attributed VMProtect-protected executables with a macro-F_1 of 91.2%, above trees on the loader-relevant fields alone (90.9%), the multi-view Transformer we develop (87.5%), and every image and byte network, and it trained in seconds per fold.
- We show that PIFA alone resists every header manipulation we test. It kept a macro-F_1 of 91.1%, 90.9%, and 91.0% under forgery, false-flag, and editing attacks, whereas clean-trained trees on all header fields, more accurate on untouched samples (92.5%), collapsed to 50.2% under a false flag, and trees on the loader-relevant fields fell to 85.3% under editing.
- We develop ObfusFormer, a multi-view Transformer with obfuscation-paired training, and build a leakage-free, build-grouped, adversary-aware benchmark from our earlier VM-obfuscated corpus. Splitting the augmented images at image level raised the accuracy of Mal-CNN on protected malware from 16.5% to 81.2%, so the released split rewards the recognition of individual executables.

A project page is available, with the code and the derived datasets to be released there upon publication.[1]

1. https://panzerjagerwang.github.io/pifa-apt/.

2 Related Works and Existing Gaps

2.1 Image and Byte Representations of Executables

Static learning-based malware analysis maps the bytes of an executable to a label without running it. Nataraj *et al.* [7] rendered malware binaries as gray-scale images and classified them with texture features, reporting 98% accuracy on 9,458 samples from 25 families. Kalash *et al.* [8] replaced the texture features with a deep CNN. Yakura *et al.* [9] introduced attention to this representation and extracted family-characteristic byte regions from the attention map. Vision Transformers for malware images added self-supervised pretraining [10] and cross-scale embedding with adaptive token masking [11]. In parallel, Raff *et al.* [16] trained a model, which they named MalConv, on the raw byte sequences of whole executables. A third line of work summarizes a whole file by its byte distribution. Saxe and Berlin [17] computed a two-dimensional histogram of byte values and local entropy and fed it, with other static features, to a deep neural network. Anderson and Roth [15] released EMBER, a benchmark of static features extracted from 1.1 million Windows executables. Its reference classifier is a gradient-boosted decision tree model [18]. A fourth line of work reads the PE header. Raff *et al.* [19] trained neural networks on the raw bytes of part of the PE header and detected malware more accurately than a model on explicitly parsed header features. Webster *et al.* [20] extracted the Rich header from 964,816 malware samples and used it to identify similar malware and builds made in different build environments.

These representations share three limitations. First, the image-based studies resize a whole file into one fixed network input, and their Transformers split that image into square patches that join bytes one image width apart in the file. Second, the byte-sequence models read a protected executable of several megabytes as one long sequence, in which the code added by the protector dominates, and the distribution features discard the byte order entirely. Third, the header studies do not separate the fields that Windows uses to load a program from descriptive fields that an adversary can rewrite. None of these studies measured which of these representations still identifies the family of an executable after virtualization.

2.2 Malware Analysis Under Virtualization Obfuscation

Virtualization obfuscation translates selected program code into bytecode for a virtual machine that is generated for, and embedded in, the protected program [6]. Sharif *et al.* [21] presented Rotalume, which recovered the bytecode syntax and semantics of malware emulated with VMProtect and Code Virtualizer from x86 execution traces. Xu *et al.* [22] simplified partially virtualized binaries with VMHunt. Li *et al.* [5] extracted 1,915 verified instruction-mapping rules from four commercial obfuscators with chosen-instruction attacks. On the learning side, Aghakhani *et al.* [12] packed 50,724 executables with nine packers, none of which was VMProtect, and reported false-positive rates of up to 83.06% when tElock, PECompact, or kkrunchy was withheld from training. Gibert *et al.* [23] investigated how packing affects static machine-learning detectors, particularly visualization-based ones, and concluded that packing limits current static systems. Han *et al.* [13] protected the executables of ten APT families with VMProtect and classified images of the protected files with Mal-CNN. For Android applications, Wu *et al.* [24] used contrastive learning to reduce the difference introduced by obfuscation. Their

IFDroid classified images of function call graphs and maintained an F_1 of 98.4% on 69,421 obfuscated samples.

Deobfuscation recovers program semantics, but each sample requires execution traces or symbolic analysis [21], [22]. Li *et al.* [5] also found three state-of-the-art deobfuscation techniques insufficient against modern commercial obfuscators. The learning studies on Windows executables treat protected samples either as a separate dataset [13] or as a held-out condition [12]. IFDroid [24] relies on function call graphs recovered by program analysis, whereas virtualization replaces the protected functions with bytecode for an embedded interpreter [6]. None of these studies trained a byte-level classifier of Windows executables with an executable and its virtualization-protected version as a positive pair.

2.3 Evaluation Protocols for Malware Learning

Evaluation protocols decide whether a reported accuracy reflects performance on unseen malware. Arp *et al.* [25] studied 30 papers from top security venues and showed how pitfalls in design and evaluation lead to unrealistic performance. Pendlebury *et al.* [26] showed that spatial and temporal bias inflated Android malware classification results. Kaufman *et al.* [27] defined leakage as the use of information about the target that should not be legitimately available during learning. Boot *et al.* [3] released 3,594 samples from 12 APT groups and compared random forests with fully connected networks on sandbox and VirusTotal reports. Gray *et al.* [4] surveyed malware authorship attribution and noted the lack of assurance about how the APT dataset of Boot *et al.* was labeled. Laurenza *et al.* [28] triaged APT malware with one-class classifiers on static features. Haddadpajouh *et al.* [29] attributed payloads of five APT families with a multi-view fuzzy consensus model and found multi-view attribution more accurate than single-view attribution.

Image datasets of APT and protected malware contain a few hundred executables per family. Han *et al.* [13] enlarged theirs with duplication and geometric augmentation. If the augmented images are divided into training and test sets image by image, images derived from one executable can fall into both sets, which is a form of the leakage defined in [27]. Detection studies on EMBER-derived data face a different pitfall. The EMBER training set contains unlabeled samples [15] that must not be merged into the benign class. On EMBER-derived data, Han *et al.* [13] also compared networks with one another rather than with the gradient-boosted reference model of the benchmark. Malware collections also contain many executables of one build, which share their header fields. A random split can place such executables in both the training and the test set. Adversaries can further modify a PE file without changing its behavior, for example by editing the DOS header or injecting content into new sections [30], [31], so header fields that a classifier relies on can be forged. To our knowledge, no published study of virtualization-protected malware has evaluated classifiers with splits grouped by source executable or by build, on builds made after the training builds, or on samples with forged, copied, or edited header fields.

3 Problem Formulation and Preliminaries

3.1 Detection and Attribution of Unseen Builds

An executable is a byte string $x = (b_1, \ldots, b_L)$ with $b_l \in \{0, \ldots, 255\}$, where L denotes the file length in bytes. A virtualization protector is a mapping $O(\cdot)$ that returns an executable $O(x)$ that behaves like x but in which selected code of x is replaced with bytecode for an embedded interpreter. We study two static tasks in which a classifier observes only the bytes of a sample and never executes, unpacks, or disassembles it. In detection, the classifier predicts $y^{\mathrm{d}} \in \{0, 1\}$ (benign or malicious) for an unseen sample. In attribution, the classifier receives x or $O(x)$ without its protection state and predicts its family $y \in \{1, \ldots, K\}$, here the APT group that produced it, where K denotes the number of families.

For attribution, the training data are

$$\mathcal{D} = \{(x_i, O(x_i), y_i)\}_{i \in \mathcal{P}} \cup \{(x_i, y_i)\}_{i \in \mathcal{U}}, \tag{1}$$

where $\mathcal{D}$ denotes the training set, y_i denotes the family of executable i, $\mathcal{P}$ denotes the indices of executables that have a protected version, and $\mathcal{U}$ denotes the indices of executables available only in plain form. Executables compiled from one build share their header fields, so a test executable whose build also occurs in training can be recognized rather than attributed. We therefore assign each plain executable x a build key $k(x)$, the tuple of the time stamp, the code size, the size of initialized data, the entry-point address, and the image size in its PE header. We partition the executables into five folds stratified by family, with all executables of one build key in one fold. For fold q, the test set contains x_i for every index i in the fold and $O(x_i)$ for every such $i \in \mathcal{P}$. Neither version of a test executable, nor any executable of the same build, enters training or model selection. To test attribution of builds made after the training builds, we also split by time. For each family, the latest builds by compile time stamp that hold about 20% of its executables form the test set. We compare three training protocols on the same folds, namely Plain (train on x_i only), VM (train on $O(x_i)$ only), and Joint (train on both), and we test every model on plain and on protected test samples.

Our threat model has a defender who trains the classifier and owns a copy of the protector, so that every training executable can be protected, and an adversary who protects a malware sample with the same protector. After protection, the adversary may rewrite every header field that does not affect how Windows loads and runs a user-mode executable, such as the time stamp, the linker version, the section names, and the Rich header (Section 3.2). The adversary may also copy these fields from malware of another group, as copied Rich headers have been used to point attribution at another group. Among the loader-relevant fields, the adversary may edit those whose values do not change how the program runs, such as informational flags and the stack and heap sizes. Such functionality-preserving edits of PE files are known from attacks on static malware classifiers [31], [32]. Changes that rebuild the section layout or move the entry point lie outside this model. The adversary knows neither the training data nor the classifier.

For attribution, we report accuracy and the macro-averaged F_1 score

$$F_1^{\mathrm{macro}} = \frac{1}{K} \sum_{c=1}^{K} \frac{2 P_c R_c}{P_c + R_c}, \tag{2}$$

where F_1^{macro} denotes the macro-averaged score, and P_c and R_c denote the precision and recall of family c, respectively. We compute it on the original test samples and on test samples whose descriptive header fields have been forged or copied from another group or whose loader-relevant fields have been edited (Section 4.4). For detection, we report accuracy, F_1, the area under the receiver operating characteristic curve (ROC-AUC), and the true-positive rate (TPR) at a false-positive rate (FPR) of 1%.

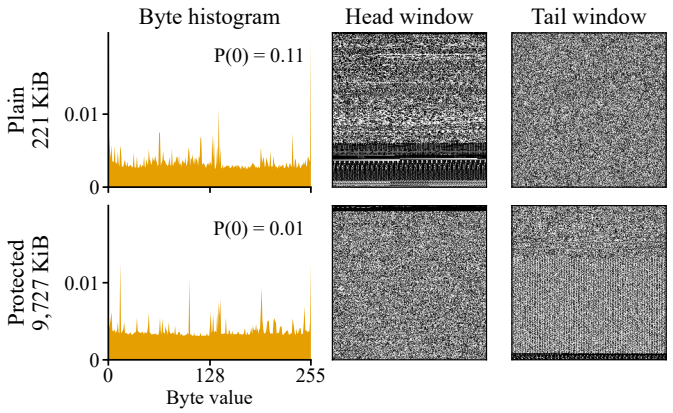


Fig. 2. Byte views of one APT29 executable before and after VMProtect protection. Columns show the byte histogram of the whole file, the head window, and the tail window; $P(0)$ is the fraction of zero bytes.

Protection changes the bytes that image and byte classifiers read (see Fig. 2). The median size ratio of a protected executable to its source was 57.3, so a protected executable of median size resized to a 128×128 image averages about 603 bytes per pixel. The PE header, in contrast, keeps much of its content (Section 5.3).

3.2 Notation, Header Fields, and Compared Methods

EMBER describes each executable by a vector of static features [15], which has 2,381 32-bit floating-point entries in feature version 2 of the EMBER code. The detection images in the archive released with [13] were produced by a renderer that reads the 9,524 bytes of this vector, repeats each byte six times, truncates the sequence at 224×224 pixels, and rescales the result to $[0, 255]$. Each image therefore contains the first 8,363 bytes of the feature vector. We invert the repetition exactly and fold these bytes, padded with zeros, into a 96×96 native image for the image networks. We also decode them into the first 2,090 feature values, from the byte histogram to 1,147 of the 1,280 hashed import features. The decoded byte histogram sums to one, as in EMBER, for 93.1% of the training images and 95.3% of the test images. In the other images, the rescaling of the renderer altered the bytes.

For attribution, we parse the PE header of every executable from its first 64 KiB. We split its fields into two groups by whether they affect how Windows loads and runs a user-mode executable. The 97 loader-relevant values describe the machine; the number of sections; the PE format; the file characteristics; the sizes of the optional header, of the image, and of the headers; the entry point; the image base; the section and file alignments; the subsystem and its version; the DLL characteristics; the stack and heap sizes; the data directories other than the certificate and debug entries; and the virtual size, virtual address, raw size, raw offset, and characteristics of the first ten sections. The 30 descriptive values, which the loader ignores, are the time stamp; the symbol-table pointer and count; the linker, operating-system, and image versions; the code and data sizes and bases; the certificate and debug directory entries; the names of the first ten sections; and the presence of a Rich header. The Rich header is a block of build-environment metadata between the DOS stub and the PE header [20]. The descriptive values also include the checksum, which Windows validates only for drivers and for DLLs loaded at boot time or into critical processes. The image networks read a global statistics view $\mathbf{G}$, which divides the whole file into $M = 256 \times 256$ cells of consecutive bytes. Its three channels hold, for each cell, the mean byte value, the Shannon entropy of the 256-byte chunk at the cell scaled by 255/8, and the fraction of printable ASCII bytes scaled by 255. The remaining symbols of the paper are defined once in Table 1.

TABLE 1
Notation

Symbol	Meaning
x, L	Executable as a byte string; its length in bytes
$O(x)$, $F(x)$	Protected version of x; x with forged descriptive fields
y^{d}, y, s	Detection label; family label; protection state
K, c	Number of families; family index
$\mathcal{P}$, $\mathcal{U}$	Indices of paired and plain-only executables
$k(x)$, q	Build key; fold index
$\mathbf{G}$, M	Global statistics view; number of its cells
$\mathbf{r}$	Structure view (97 loader-relevant header fields)
$\mathbf{d}$	Byte-distribution view (514 values)
$\mathbf{H}$, $\mathbf{T}$, W	Head window; tail window; window length in bytes
N, N_v, D	Tokens per window; tokens of value view v; token dimension
$\mathbf{z}$, $\tilde{\mathbf{z}}$	Projected embeddings of x and $O(x)$ in the paired loss
τ, α, β, ρ	Temperature; state weight; pair weight; forging probability

We compare the gradient-boosted trees of Section 4.2 with eight baseline networks, the multi-view Transformer ObfusFormer of Section 4.3, and a nearest-neighbor rule. We reimplemented the baseline CNN and Mal-CNN of Han *et al.* [13] from the layer table of their paper. The CNN stacks four 3×3 convolution blocks with batch normalization and max pooling before a 1,024-unit fully connected layer. Mal-CNN replaces each convolution with a depth-wise separable convolution and adds a 7×7 spatial attention gate. In attribution, both read the 128×128 contrast-enhanced images released with that study. Our MalConv [16] reads the concatenated head and tail windows, embeds each byte, applies gated convolutions with kernel size and stride 512, and max-pools over time. The ImageNet-pretrained [33] CNNs ResNet-18 [34], MobileNetV2 [35], and EfficientNet-B0 [36] and Transformers DeiT-Ti [37] (hereafter ViT-Ti) and Swin-T [38] read the global statistics view in attribution. MobileNetV2 and EfficientNet-B0 enter only the detection experiment, and MalConv only the attribution experiment. The nearest-neighbor rule assigns a test sample the family of the training sample with the fewest differing loader-relevant field values. We keep, for each run, the epoch with the highest validation accuracy in detection, or the highest validation macro-F_1 (ties broken by validation accuracy) in attribution, and, for the trees, the boosting round with the lowest validation loss.

4 Method

Whole-file images average away the fine structure of a protected file, and byte sequences mix the regions that protection rewrites with those it keeps. The PE header, in contrast, keeps the family signal, but part of it lies in descriptive fields that an adversary can forge or copy from another group. PIFA therefore reads every header field with gradient-boosted trees, yet trains them on forged, copied, and edited copies of each executable, so that it uses a field only where the field agrees with the loader-relevant structure that an adversary cannot change. To test whether a learned representation of the raw file can match PIFA, we develop the Obfuscation-Paired Multi-View Transformer (ObfusFormer), which reads the loader-relevant fields, the byte distribution of a whole file, and the two ends of the file through view-specific tokenizers, and which we train on each executable together with

TABLE 2
Datasets. Attribution counts are executables; each protected executable has a plain source, and builds are distinct build keys. *Unlabeled EMBER images stored in the benign folder, removed. †One build, left out

Detection (EMBER-derived)			Attribution (APT)			
Split	Benign	Malicious	Family	Plain	Prot.	Builds
Train	20,524	20,721	APT1	404	399	269
Test	2,454	2,546	APT10	228	223	213
Unlabeled*	13,755	–	APT28	162	160	154
			APT29	263	251	199
			APT30	164	157	145
			DarkHotel	269	260	218
			Energetic Bear	132	127	82
			Equation Group†	395	387	1
			Gorgon Group	345	284	323
			Winnti	385	326	287
Total	22,978	23,267	Used	2,352	2,187	1,890

its protected version and on copies with manipulated header fields. The datasets and splits come first (Section 4.1), followed by PIFA (Section 4.2), the Transformer (Section 4.3), and the training and test samples with paired and manipulated headers (Section 4.4).

4.1 Detection and Attribution Datasets Grouped by Build

We build one dataset for each task from the archive released with the study of Han *et al.* [13] (Table 2). The training split of the detection dataset, which contains EMBER-derived byte images [15], holds 55,000 images, of which 13,755 come from unlabeled EMBER samples that the original scripts stored in the benign folder. We remove these unlabeled images, which leaves 20,524 benign and 20,721 malicious training images. The test split of 2,454 benign and 2,546 malicious images, which the original scripts drew from the EMBER test set, stays unchanged. A fixed random 10% of the training images forms the validation set.

The attribution dataset contains APT malware whose files are named by their SHA-256 hashes. The per-group file counts match the public collection of Boot *et al.* [3] except for one APT1 file. Han *et al.* studied ten of its groups and protected their PE files with VMProtect. We pair each protected file with its plain source through the SHA-256 hash of the source, which gives 2,747 plain and 2,574 protected executables. These executables have only 1,891 distinct build keys. All 395 Equation Group executables share one build key and one file size of 184,320 bytes, so a split grouped by build would place the whole group in one fold. We therefore exclude the Equation Group from the attribution experiments, which leaves 2,352 plain executables of $K = 9$ groups with 1,890 build keys, $|\mathcal{P}| = 2{,}187$, and $|\mathcal{U}| = 165$. We split them into five folds as defined in Section 3.1. In each fold, whole builds that hold about 10% of the training executables of each family form the validation set. On the split by time, the latest builds that hold about 10% of the remaining executables of each family form the validation set, and executables whose time stamp lies outside 2000 to 2020 always stay in training. Its 474 test executables, 404 of them with a protected version, were compiled from 2012 to 2018. No test build of a family is older than its training and validation builds with a valid time stamp.

4.2 PIFA: Adversarially Trained Header Attribution

The descriptive header fields carry a real family signal. Trees that read all header fields attribute protected executables more accurately than trees on the loader-relevant fields alone (Section 5.3). An adversary, however, can forge those fields or copy them from another group, which collapses such a classifier. Restricting a classifier to the loader-relevant fields, which virtualization preserves and which cannot be copied from another group without changing how a program runs, removes that vulnerability but discards the descriptive signal and still leaves the loader-relevant fields open to functionality-preserving edits. PIFA instead reads every header field with gradient-boosted decision trees [18], but trains them to resist the manipulations an adversary can apply, so that a field is used only to the extent that it agrees with the non-forgeable loader-relevant structure. For every training executable we add three adversarial copies, each labeled with its true family, namely a forged copy with random descriptive fields (Section 4.4), a false-flag copy that takes the descriptive fields of a random training executable of another family (the attack of Section 3.1), and an edited copy with functionality-preserving changes of the loader-relevant fields. A split on a forged or copied field then gains nothing, so training drives the trees to trust a descriptive field only where it agrees with the structure an adversary cannot change. The trees learn no features and train in seconds per fold on the laptop CPU. We compare PIFA with trees that read, without this adversarial training, all header fields, the loader-relevant fields alone, the descriptive fields alone, and the byte-distribution view of Section 4.3. In detection, the trees read the 2,090 decoded feature values with the hyperparameters of the EMBER reference model [15].

4.3 Multi-View Transformer

ObfusFormer renders each executable into up to four views, tokenizes each view, and fuses the tokens of all views with one Transformer encoder (see Fig. 3). The structure view $\mathbf{r} \in \mathbb{R}^{97}$ holds the loader-relevant header fields of Section 3.2, with zeros for absent sections. The byte-distribution view $\mathbf{d}$ summarizes the whole file without regard to byte order:

$$\mathbf{d} = [\mathbf{q}^{\mathrm{b}};\, \mathbf{q}^{\mathrm{e}};\, \log_2 L;\, \bar{e}_{\mathrm{f}}] \in \mathbb{R}^{514}, \tag{3}$$

where $\mathbf{q}^{\mathrm{b}} \in \mathbb{R}^{256}$ and $\mathbf{q}^{\mathrm{e}} \in \mathbb{R}^{256}$ denote the byte histogram and the byte-entropy histogram of the file, both divided by L, and $\bar{e}_{\mathrm{f}}$ denotes the mean Shannon entropy of its 256-byte chunks. The byte-entropy histogram counts the bytes of each chunk in a 16×16 table of its entropy bin and the high nibble of the byte value, in the manner of Saxe and Berlin [17]. The head window $\mathbf{H}$ holds the first $W = 65{,}536$ bytes (the PE header, the section table, and the start of the sections) and the tail window $\mathbf{T}$ the last W bytes, both zero-padded as a 256×256 image.

A value tokenizer turns a value view $v \in \{\mathbf{r}, \mathbf{d}\}$ into N_v tokens. Because header fields and histogram entries range over several orders of magnitude, it first compresses each value a to $\mathrm{sign}(a)\log(1+|a|)$ and zero-pads the view to $16N_v$ values. For the structure view, it then encodes each value piecewise linearly over 16 bins whose edges are the quantiles of that field in the training samples [39], hereafter the bin encoding; for the byte-distribution view, it standardizes the values with batch normalization. It splits the encoded values into groups of 16 consecutive values and maps each group with its own linear layer:

$$\mathbf{E}_{v,g} = \mathbf{W}_{v,g}\tilde{\mathbf{v}}_g + \mathbf{b}_{v,g}, \quad g = 1, \ldots, N_v, \tag{4}$$

where $\mathbf{E}_{v,g} \in \mathbb{R}^D$ denotes token g of view v, $\tilde{\mathbf{v}}_g$ the encoded values of group g, and $\mathbf{W}_{v,g}$, $\mathbf{b}_{v,g}$ its weights and bias; the 97 and 514 values give $N_{\mathbf{r}} = 7$ and $N_{\mathbf{d}} = 33$ tokens.

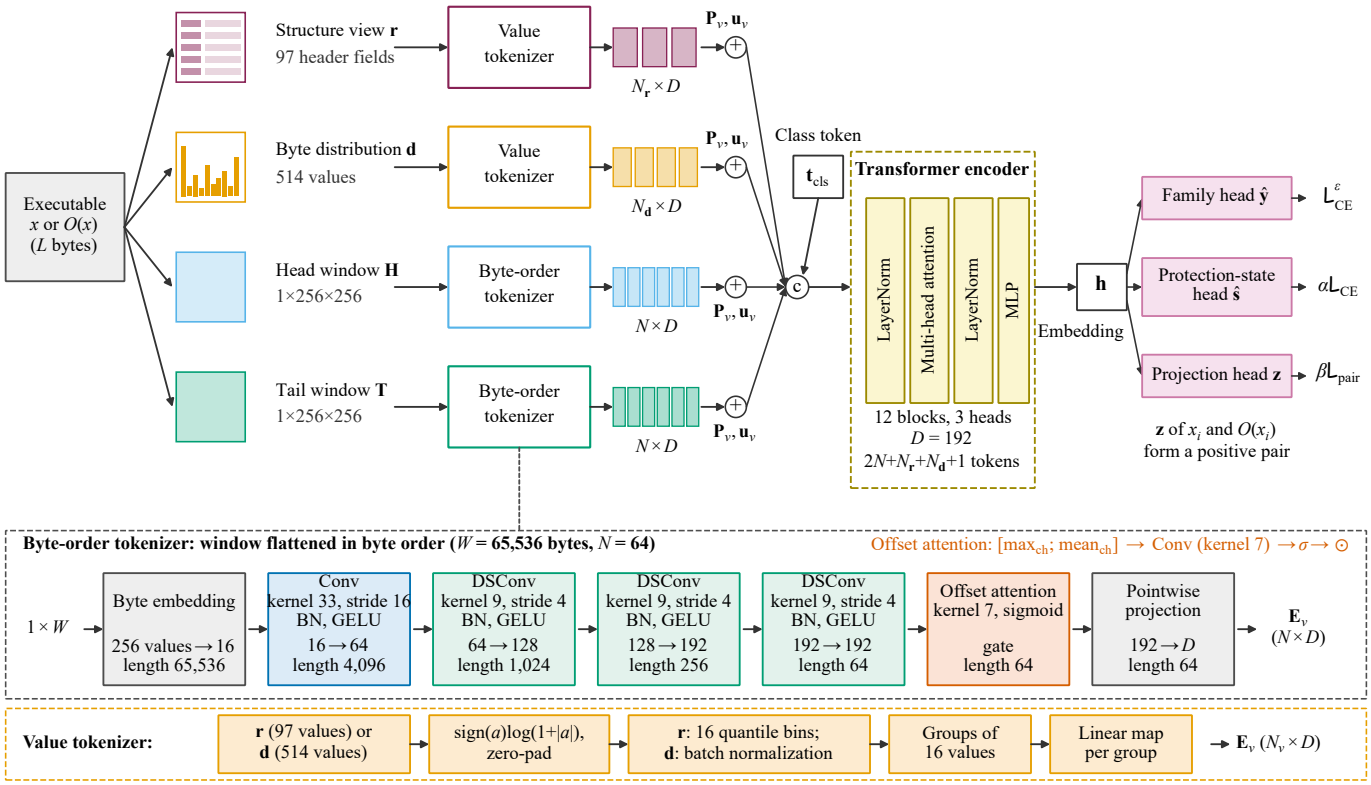


Fig. 3. Architecture of ObfusFormer. Value tokenizers embed the structure view and the byte-distribution view, and each window is flattened in byte order and converted into N tokens by its own byte-order tokenizer. The tokens of all views, with positional and view embeddings, enter a shared Transformer encoder. The class-token embedding $\mathbf{h}$ feeds the family head, the protection-state head, and the projection head of the obfuscation-paired loss. The zoomed panels show one byte-order tokenizer and the value tokenizer.

A square image patch joins bytes one image width apart, whereas executable bytes follow their order. The byte-order tokenizer therefore flattens each window into a sequence of W bytes, embeds each byte value as $\boldsymbol{\phi}_b \in \mathbb{R}^{16}$ to give the input $\mathbf{X}_v \in \mathbb{R}^{16\times W}$ of window v, and maps it to N tokens of consecutive bytes:

$$\mathbf{F}_v = \mathcal{S}_R \circ \cdots \circ \mathcal{S}_1\big(\mathrm{GELU}(\mathrm{BN}(\mathrm{Conv}_{r_0}(\mathbf{X}_v)))\big), \tag{5}$$

where Conv_{r_0} is a one-dimensional convolution with stride $r_0 = 16$ and kernel $2r_0 + 1$, BN is batch normalization, GELU is the Gaussian error linear unit, and each of the $R = 3$ stages $\mathcal{S}_r$ is a depth-wise separable convolution (adapted from Mal-CNN [13]) with stride 4 and kernel 9, then BN and GELU. With $N = 64$, each token summarizes 1,024 consecutive bytes. An offset attention gate adapts its spatial attention to the offset axis. It multiplies $\mathbf{F}_v$ at each offset by the sigmoid of a kernel-7 convolution over the channel-wise maximum and mean, and a pointwise projection then gives the token sequence $\mathbf{E}_v \in \mathbb{R}^{N\times D}$ of window v.

We add to every token a learned positional embedding $\mathbf{P}_v$ of its view and a learned view embedding $\mathbf{u}_v \in \mathbb{R}^D$, and we prepend a class token $\mathbf{t}_{\mathrm{cls}}$:

$$\mathbf{S}^{(0)} = [\mathbf{t}_{\mathrm{cls}};\ \mathbf{E}_{v_1} + \mathbf{P}_{v_1} + \mathbf{u}_{v_1};\ \ldots;\ \mathbf{E}_{v_V} + \mathbf{P}_{v_V} + \mathbf{u}_{v_V}], \tag{6}$$

where $\mathbf{S}^{(0)}$ denotes the input sequence of the encoder and $v_1, \ldots, v_V$ denote the V views that a configuration reads. The encoder has the structure of ViT-Ti (DeiT-Ti [37]). It stacks 12 pre-norm blocks of three-head self-attention [40] and a multilayer perceptron, with $D = 192$, initialized from ImageNet-pretrained ViT-Ti while the tokenizers and embeddings train from scratch. The normalized class token after the last block gives the embedding $\mathbf{h} \in \mathbb{R}^D$, which feeds a linear family head $\hat{\mathbf{y}} = \mathrm{softmax}(\mathbf{W}_y\mathbf{h}+\mathbf{b}_y)$, a linear protection-state head $\hat{\mathbf{s}}$, and a two-layer projection head $\mathbf{z} = \pi(\mathbf{h})$ used only during training. For detection, ObfusFormer reads the 2,090 decoded feature values as one value view with $N_v = 131$ tokens, and the family head becomes a two-class detection head.

4.4 Paired and Manipulated Samples

A protected executable and its plain source carry the same family label but share few byte statistics. We treat each training pair as two versions of one executable and pull their projected embeddings together, in the manner of contrastive learning on augmented views [14] with an InfoNCE loss [41]. Each mini-batch contains B executables drawn without replacement. Every executable i in $\mathcal{P}$ enters the batch with both x_i and $O(x_i)$, and every executable in $\mathcal{U}$ enters with x_i only. The obfuscation-paired loss over the batch is

$$\mathcal{L}_{\mathrm{pair}} = -\frac{1}{2B_p}\sum_{i=1}^{B_p}\left[\log\frac{\exp(\langle \mathbf{z}_i, \tilde{\mathbf{z}}_i\rangle/\tau)}{\sum_j \exp(\langle \mathbf{z}_i, \tilde{\mathbf{z}}_j\rangle/\tau)} + \log\frac{\exp(\langle \tilde{\mathbf{z}}_i, \mathbf{z}_i\rangle/\tau)}{\sum_j \exp(\langle \tilde{\mathbf{z}}_i, \mathbf{z}_j\rangle/\tau)}\right], \tag{7}$$

where $\mathcal{L}_{\mathrm{pair}}$ denotes the paired loss, $B_p \le B$ denotes the number of executables of the batch that have a protected version, $\mathbf{z}_i$ and $\tilde{\mathbf{z}}_i$ denote the ℓ_2-normalized projections of x_i and $O(x_i)$, respectively, $\langle\cdot,\cdot\rangle$ denotes the inner product, j runs over the B_p pairs of the batch, and τ denotes the temperature. The other pairs of the batch, including pairs of the same family, act as negatives. The full objective is

$$\mathcal{L} = \mathcal{L}^{\epsilon}_{\mathrm{CE}}(\hat{\mathbf{y}}, y) + \alpha\,\mathcal{L}_{\mathrm{CE}}(\hat{\mathbf{s}}, s) + \beta\,\mathcal{L}_{\mathrm{pair}}, \tag{8}$$

where $\mathcal{L}$ denotes the training loss, y and s denote the family and protection state of a sample, $\mathcal{L}_{\mathrm{CE}}^{\epsilon}$ denotes cross-entropy with label smoothing ϵ [42], $\mathcal{L}_{\mathrm{CE}}$ denotes cross-entropy, both averaged over the samples of the batch, α weights the protection-state loss, and β weights the paired loss. Under the Plain and VM protocols, which have no pairs and one protection state, we set $\alpha = \beta = 0$.

The head window holds the descriptive header fields at fixed offsets, so a byte-order tokenizer can learn their values. To keep the configurations of ObfusFormer that read a window from relying on them, we train these configurations also on forged copies. The forging function $F(\cdot)$ draws random values for every descriptive field, namely the time stamp, the linker, operating-system, and image versions, the checksum, the code and data sizes and bases, and the names of the first ten sections. It zeroes the symbol table, the certificate and debug directory entries, and the Rich header. It leaves every byte of a loader-relevant field unchanged, so $F(x)$ runs like x and has the same structure view. All changed bytes lie in the first 1,088 bytes of a file. We therefore update the other views exactly, namely the byte histogram, the byte-entropy histogram, and the mean chunk entropy from the changed 256-byte chunks, the head window directly, and, for the image networks, the cells of $\mathbf{G}$ that cover the changed bytes. We precompute eight forged copies of every executable, with different random seeds, and replace each training sample with a randomly chosen copy with probability ρ. A ninth copy, with another seed, forms the forged test set of Section 3.1. Algorithm 1 summarizes the training procedure.

Two further manipulations change the parsed header fields rather than the bytes, so they apply only to the methods that read header fields. The editing function toggles, each with probability 0.5, the six characteristics flags that the PE specification marks as deprecated or informational for images and the terminal-server-aware DLL flag. It also sets the minor subsystem version to a random value from 0 to 9 when the major version is below 10, draws new stack and heap reserve sizes (64 KiB to 4 MiB) and commit sizes (4 KiB to 64 KiB), and grows the virtual size of the last listed section within its section alignment, so that the image size stays unchanged. As with forging, we precompute nine edited copies of every executable, test on the ninth, and train ObfusFormer on the other eight. The trees train on the original samples together with one edited copy of each. For a false flag, every protected test executable receives all descriptive fields of a random protected training executable of another group.

Algorithm 1 Training of ObfusFormer with paired and forged samples

Require: Training indices $\mathcal{P} \cup \mathcal{U}$, executables x_i, protected versions $O(x_i)$, families y_i, protection states s_i, forged copies $F^{(1)}(\cdot), \ldots, F^{(8)}(\cdot)$; validation set; epochs E; probability ρ
Ensure: Trained parameters θ^*
Initialize the encoder from ViT-Ti and the tokenizers, embeddings, and heads randomly; fit the bin edges of the structure view on the training samples
for $t = 1$ **to** E **do**
  for each batch $\mathcal{B}$ of B indices from $\mathcal{P} \cup \mathcal{U}$ **do**
    Collect x_i for $i \in \mathcal{B}$ and $O(x_i)$ for $i \in \mathcal{B} \cap \mathcal{P}$
    Replace each sample with a random forged copy with probability ρ
    Render the views of the configuration: **r**, **d** with (3), and the windows **H**, **T**
    Tokenize **r** and **d** with (4) and the windows with (5) and the offset attention gate
    Encode the sequence (6) and read **h**, $\hat{\mathbf{y}}$, $\hat{\mathbf{s}}$, **z**
    Compute $\mathcal{L}_{\mathrm{pair}}$ over the pairs in $\mathcal{B}$ with (7)
    Update θ with AdamW on $\mathcal{L}$ in (8)
  end for
  Compute the validation macro-F_1 with (2); if it is the best so far, set $\theta^* \leftarrow \theta$
end for
return θ^*

TABLE 3
Training hyperparameters. †ViT-Ti, Swin-T, EfficientNet-B0, and the encoder of ObfusFormer

Item	Detection	Attribution
Epochs	8	15
Batch (samples)	32	16 (Joint: 8 executables)
Learning rate	3×10^{-4}	3×10^{-4}
CNN, Mal-CNN, MalConv	5×10^{-4}	5×10^{-4}
Weight decay, stochastic depth†	0.05, 0.1	0.05, 0.1
Label smoothing ϵ	0	0.1
ObfusFormer value tokens	131	7 (**r**), 33 (**d**)
ObfusFormer window tokens N, bins of **r**	–	64, 16
τ, α, β, ρ	–	0.1, 0.2, 0.5, 0.5
Swin-T gradient-accumulation steps	1	4

5 Experiment

5.1 Experimental Setup

We ran the experiments on a laptop with an Intel Core i5-12450H CPU, 16 GB of memory, and an NVIDIA GeForce RTX 4050 Laptop GPU with 6 GB of memory, under Windows 11, with Python 3.11.15, PyTorch 2.11.0 with CUDA 12.6, timm 1.0.30, scikit-learn 1.8.0, and LightGBM 4.6.0. A second machine had an Intel Core i7-14650HX CPU and an NVIDIA GeForce RTX 4070 Ti SUPER GPU with 16 GB of memory and ran Ubuntu 24.04, Python 3.12.13, PyTorch 2.11.0 with CUDA 12.8, and the same timm version. On it, we ran the attribution runs of the networks and of configurations of ObfusFormer that read more than the structure view, and the detection runs of all networks with seed 2 and of MobileNetV2 and EfficientNet-B0 with seed 1. We ran the detection runs of Mal-CNN and CNN with seed 0 and the Mal-CNN runs on the released augmented images on the CPU in 32-bit precision, and all trees on the CPU of the laptop. We trained all networks with shared hyperparameters and used method-specific values only where marked (Table 3). On the GPU, we used bfloat16 autocasting. The optimizer was AdamW [43] with a cosine learning-rate schedule, one warm-up epoch, and gradient-norm clipping at 1.0. The trees in detection used the hyperparameters of the EMBER reference model [15]. In attribution, the trees had 31 leaves, at least 10 samples per leaf, a feature fraction of 0.5, and per-round bagging of 80%, with boosting stopped after 100 of at most 2,000 rounds without an improvement of the validation loss.

We selected the views of ObfusFormer on the validation splits of the Joint protocol (Section 5.4). Among the six view configurations, the structure view alone had the highest mean validation macro-F_1, 88.3%, against 79.0% for the byte-distribution view and the two windows. In the comparisons, ObfusFormer denotes this configuration, and ObfusFormer-B denotes the configuration with the byte-distribution view and the two windows, which reads the same inputs as the byte-level methods. For detection, batch-normalized value tokens had the highest validation accuracy among five tokenizations with seed 0, 94.3%, against 93.6% for the bin encoding.

TABLE 4
Detection on the EMBER-derived test images: means over three seeds. The trees and ObfusFormer read the decoded feature values, and the other networks read the native images. The best value in each column is in bold

Method	Acc. (%)	F_1 (%)	AUC	TPR@1% (%)
Networks of Han et al. [13]				
CNN	90.9	91.3	0.971	70.0
Mal-CNN	88.5	88.8	0.957	62.9
ImageNet-pretrained networks				
MobileNetV2	89.4	89.7	0.960	64.5
ResNet-18	89.6	90.0	0.963	63.2
EfficientNet-B0	90.1	90.4	0.964	61.8
ViT-Ti	90.2	90.5	0.965	65.9
Swin-T	90.7	90.9	0.970	68.7
Gradient-boosted trees				
Decoded feature values	**94.7**	**94.8**	**0.988**	**82.9**
Transformer (this work)				
ObfusFormer	91.5	91.7	0.971	72.0

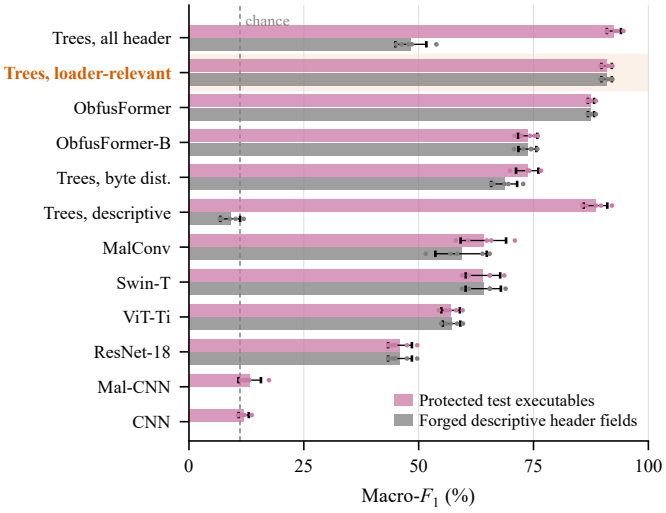


Fig. 4. Attribution under protection. Macro-F_1 on the protected test executables under the Joint protocol on the folds grouped by build (seed 0), on the original executables and on executables with forged descriptive header fields. The bars are means over the five folds, the caps the standard deviation, and the dots the folds. The dashed line marks the macro-F_1 of a uniformly random classifier of nine families. The CNN and Mal-CNN read released images that cannot be re-rendered and have no forged result; the trees on the loader-relevant fields are highlighted.

5.2 Experiment 1: Malware Detection

We trained each detection method with three seeds on the cleaned training images and evaluated it on the 5,000 test images (Table 4). The gradient-boosted trees on the decoded feature values detected malware most accurately, with an accuracy of 94.7% and a ROC-AUC of 0.988. ObfusFormer had an accuracy of 91.5%, 3.2 points below the trees and 0.6 points above CNN, the most accurate image network (90.9%). With the unlabeled samples kept in the benign class, as in the original scripts, Mal-CNN had an accuracy of 86.3% with seed 0, against 88.1% on the cleaned images.

5.3 Experiment 2: Attribution Under VMProtect

We first measured how the released split of the protected-image dataset affected accuracy. Over three seeds, Mal-CNN had a mean accuracy of 81.2% on the released image-level split, in which 820 of 1,015 test executables also contributed images to training. On a split of the same images grouped by source executable, with the same training settings, the accuracy of our re-implementation fell to 16.5% and its macro-F_1 to 18.3%. One executable contributed up to 613 augmented images to the released set.

VMProtect kept much of the PE header of its input. In all 2,574 pairs, the protected file had the time stamp, the linker version, the image base, the section and file alignments, the subsystem, the DLL characteristics, and the code size of its source. VMProtect changed the entry point, the image size, and the number of sections of every file, adding a median of three sections. The section table of 1,966 protected files (76.4%) still listed every section name of the source. VMProtect removed the Rich header, which 2,387 of the 2,747 plain executables contained. Builds also repeat. 1,095 of the 2,747 plain executables share their build key with another executable. With folds stratified only by family, 1,020 of the 2,747 plain executables had their build in the training part of another fold. We therefore report all other attribution results on the folds grouped by build and on the split by time.

On unseen builds, the trees on all header fields attributed protected test executables most accurately, with a macro-F_1 of 92.5% under the Joint protocol, and the trees on the loader-relevant fields alone with 90.9% (Table 5); we make these fields robust to manipulation below. ObfusFormer, which reads the same loader-relevant fields, had 87.5%, below both; the loader-relevant trees were more accurate than ObfusFormer in five of the five folds. The nearest-neighbor rule on these fields had 85.9%. The methods that read images or bytes ranked lower. The byte-view configuration ObfusFormer-B had 73.7%, and the baseline networks at most 64.1% (MalConv), down to 11.9% for the CNN of Han *et al.* (see Fig. 4). A bootstrap over the 2,187 protected test executables (10,000 paired, family-stratified resamples with the seed-0 predictions) confirmed both gaps. The trees exceeded ObfusFormer by 3.4 points (95% confidence interval [2.3, 4.5]), and ObfusFormer exceeded the most accurate baseline network by 23.0 points ([20.9, 25.1]). ObfusFormer has 5.7M parameters and trained in 2.0 minutes per fold on the laptop GPU, whereas the header-field trees trained in 3 seconds per fold on its CPU, so they were both the most accurate and the cheapest.

The family signal of the loader-relevant fields lay mainly in the layout of a program. The section table and the data directories dominated. Permutation importance lowered the protected-test macro-F_1 by 49.5 and 28.9 points when they were shuffled, far more than the toolchain fields (the machine, the alignments, the image base, and the stack and heap sizes) or the coarse sizes. The trees on the layout fields alone had a protected-test macro-F_1 of 90.0%, against 58.3% on the toolchain fields alone, and 80.1% on the coarse file sizes and section count alone, so the surviving signal is largely a size fingerprint of related builds. Trained on plain executables alone, the trees on the loader-relevant fields reached only 16.3% on protected ones, and every network at most 10.5%; restricted to the 49–52 fields that VMProtect kept in at least 95% of the training pairs, they recovered to 89.1%.

Forging the descriptive header fields of the protected test executables lowered the macro-F_1 of the trees on all header fields from 92.5% to 48.3% (Table 6). The trees on the descriptive fields alone fell from 88.5% to 9.0%, close to the 11% of a uniformly random classifier of nine families. The forged bytes also lowered the byte-level methods that read the head window, namely MalConv (from 64.1% to 59.2%), the byte-distribution trees, and ObfusFormer-B trained without forged copies. ObfusFormer-B

TABLE 5
Attribution macro-F_1 (%) of APT executables, training protocol → test version: means ± standard deviations over five folds (seed 0), 15 runs (seeds 0–2, grouped folds), or three runs (later builds). ∗: forged descriptive header fields, which cannot be re-rendered in the released images of CNN and Mal-CNN. The best value in each column is in bold; PIFA adds adversarial training to the all-header trees, and its robustness to header manipulation is in Table 6

	Folds grouped by build						Later builds	
	Seed 0					Seeds 0–2	Seeds 0–2	
Method	Plain→VM	VM→VM	Joint→Plain	Joint→VM	Joint→VM∗	Joint→VM	Joint→Plain	Joint→VM
Networks of Han et al. [13]								
CNN	5.6±2.1	15.2±3.8	75.0±3.3	11.9±1.1	–	12.0±1.1	45.3±1.3	13.6±0.5
Mal-CNN	5.0±1.3	16.9±4.0	73.4±3.2	13.2±2.4	–	14.5±3.0	43.1±1.2	12.7±2.8
Other networks								
MalConv	7.8±1.6	61.5±5.8	82.3±2.3	64.1±4.9	59.2±5.6	64.2±3.9	51.3±2.3	49.7±1.5
ResNet-18	3.9±0.5	41.5±1.4	83.7±2.2	45.9±2.6	45.9±2.6	46.1±2.3	50.9±1.0	34.5±0.6
ViT-Ti	6.0±1.6	58.3±2.4	81.5±2.4	56.9±2.0	57.2±1.9	57.9±3.5	50.9±2.0	40.3±3.1
Swin-T	5.5±1.1	62.9±2.5	84.1±2.2	64.0±3.8	64.1±3.8	62.7±3.3	53.2±4.5	45.3±1.6
Gradient-boosted trees								
Loader-relevant fields	16.3±3.4	89.5±1.1	91.0±1.5	90.9±1.1	**90.9±1.1**	90.6±1.3	**59.6±0.3**	**61.0±0.6**
All header fields	43.6±4.2	**92.4±1.1**	**92.8±1.5**	**92.5±1.5**	48.3±3.4	**92.5±1.7**	50.3±1.2	52.2±0.6
Descriptive fields	**65.2±1.8**	87.9±2.2	89.1±3.3	88.5±2.6	9.0±2.2	88.4±2.5	27.8±3.0	22.5±3.7
Byte distribution	4.6±1.8	74.0±3.8	88.3±1.2	73.6±2.4	68.6±2.8	72.9±2.7	56.0±0.4	46.3±0.3
Nearest neighbor								
Loader-relevant fields	–	–	88.5±0.9	85.9±2.8	85.9±2.8	–	55.2	56.0
Transformer (this work)								
ObfusFormer	10.5±7.3	86.9±1.5	87.6±1.5	87.5±0.7	87.5±0.7	87.3±1.4	57.3±0.8	57.8±1.8
ObfusFormer-B	8.2±3.0	60.8±3.7	80.9±3.8	73.7±2.1	73.7±2.0	70.0±8.1	52.9±6.2	50.1±4.3

TABLE 6
Header manipulations of the protected test executables: macro-F_1 (%) under the Joint protocol on folds grouped by build (seed 0, five folds) and the share attributed to the donor group of a false flag (%). PIFA reads all header fields and is trained on the forged, edited, and false-flag copies; the best robust value in each column is in bold

Method	Original	Forged	Edited	False flag	Donor
Gradient-boosted trees					
PIFA (all fields, adversarial)	**91.2**	**91.1**	**91.0**	**90.9**	**1.6**
Loader-relevant fields	90.9	90.9	85.3	90.9	–
Train also on edited copies.	90.8	90.8	90.5	90.8	–
All header fields	92.5	48.3	91.8	50.2	29.7
Descriptive fields	88.5	9.0	88.5	0.0	100.0
Nearest neighbor					
Loader-relevant fields	85.9	85.9	84.6	85.9	–
Transformer (this work)					
ObfusFormer	87.5	87.5	68.3	87.5	–
Train also on edited copies.	87.2	87.2	86.8	87.2	–

trained with forged copies instead kept 73.7% on them, and the image networks changed by at most 0.3 points, because forging changes only the first 1,088 bytes.

False flags redirected the attribution of the trees that read descriptive fields. With the descriptive fields of a training executable of another group, the trees on all header fields had a macro-F_1 of 50.2% and attributed 29.7% of the protected test executables to the group of the donor. Edits of loader-relevant fields lowered the macro-F_1 of the methods that read these fields. The trees on the loader-relevant fields fell from 90.9% to 85.3%, the nearest-neighbor rule from 85.9% to 84.6%, and ObfusFormer by 19.2 points. Training on edited copies limited the loss to 0.3 points for the loader-relevant trees and 0.4 points for ObfusFormer.

PIFA, which reads all header fields but trains on forged, copied, and edited copies of every executable, resisted all three manipulations. It attributed the untouched protected test executables with a macro-F_1 of 91.2% (91.1% over seeds 0 to 2), above the trees on the loader-relevant fields alone, and kept 91.1%, 90.9%, and 91.0% under forging, a false flag, and editing, sending only 1.6% of the false-flagged samples to the donor group (Table 6). No other method stayed accurate under all three attacks. The all-header trees collapsed under forging and false flags, and the loader-relevant trees fell to 85.3% under editing. PIFA's worst case over the three attacks, 90.9%, is thus the highest of any method, which makes PIFA the state of the art for robust attribution under this threat model.

On the split by time, the methods attributed the later builds less accurately than the unseen builds of the grouped folds, except for the CNN of Han *et al.*, which stayed near the 11% of a random classifier on both (Table 5). The trees on the loader-relevant fields had a protected-test macro-F_1 of 61.0%, 29.6 points below their grouped-fold value, and PIFA reached 58.3%, slightly below them. ObfusFormer had 57.8%, the trees on all header fields 52.2%, and the most accurate baseline network, MalConv, 49.7%. The nearest-neighbor rule had 56.0%, or 65.4% with the validation builds, the latest before the test builds, added to its references.

5.4 Ablation of ObfusFormer

We changed one design choice of ObfusFormer at a time, retrained it under the Joint protocol on all five folds, and compared each variant with the configuration it was derived from on the same folds and seeds (Table 7). The structure view alone gave the highest protected-test macro-F_1, 87.5% with seed 0; adding the two windows lowered it to 78.7% and the windows and byte distribution to 77.9%, and without the structure view the byte views reached at most 73.7%. Over seeds 0 to 2, the bin encoding and the ImageNet initialization mattered. Standardization instead of the bin encoding changed the protected-test macro-F_1 of ObfusFormer by −7.3 points and random initialization by −1.7, whereas removing the paired loss or the protection-state head changed it by

TABLE 7
Views and training of ObfusFormer under the Joint protocol on folds grouped by build: macro-F_1 (%) on the validation splits and on plain, protected, and forged protected ($*$) test executables with seed 0 (means over five folds), and on protected test executables over the 15 runs of seeds 0–2 (mean ± standard deviation). Configurations that read a window train with forged copies

	Seed 0				Seeds 0–2
Configuration	Val.	Plain	Prot.	Prot.$*$	Prot.
Views					
r	88.3	87.6	87.5	87.5	87.3±1.4
d	57.0	84.5	16.9	16.8	16.9±3.4
H, **T**	72.1	72.5	67.6	67.4	68.8±7.2
d, **H**, **T**	79.0	80.9	73.7	73.7	70.0±8.1
r, **H**, **T**	83.7	82.6	78.7	78.7	76.5±2.8
r, **d**, **H**, **T**	83.1	83.9	77.9	77.9	77.4±3.4
Training					
ObfusFormer	88.3	87.6	87.5	87.5	87.3±1.4
Remove the paired loss.	89.0	88.5	87.9	87.9	87.5±1.3
Remove the protection-state head.	87.3	86.5	86.7	86.7	86.8±1.8
Replace the bins with standardization.	81.3	81.5	80.0	80.0	80.0±1.6
Initialize the encoder randomly.	86.7	85.8	85.3	85.3	85.6±2.5
ObfusFormer-B	79.0	80.9	73.7	73.7	70.0±8.1
Remove the forged copies.	78.3	80.7	71.2	61.0	72.0±5.5
Remove the paired loss.	73.0	77.5	65.0	64.9	63.3±6.0
Remove the protection-state head.	73.5	75.7	64.6	64.3	68.3±4.8

+0.2 and −0.5 points, less than the seed band. For ObfusFormer-B, the forged copies helped only on forged samples (−9.6 points).

6 Discussion

6.1 What Survives Virtualization

PIFA works because the family signal lies mainly in the parts of the PE header that virtualization preserves. VMProtect protected the code of a program but kept the original section table and data directories in the header, even as it appended sections and moved the entry point. These fields carried 81.9% of the split gain of the trees on the loader-relevant fields, so the surviving signal is largely how a group lays out its programs. This signal is in part a fingerprint of related builds. On builds compiled after the training builds, the trees were 29.6 points less accurate than on the folds grouped by build, and the nearest-neighbor rule on the same fields came within 1.6 points of ObfusFormer and improved when the latest earlier builds were added to its references. PIFA therefore attributes a new sample most reliably when related builds of its group are available, and a split that ignores builds overstates this accuracy.

The descriptive header fields add a real but forgeable signal. Trees that read them were more accurate than the loader-relevant trees on untouched samples, but forging the descriptive fields or copying them from another group lowered their macro-F_1 by more than 40 points, and with the copied fields they sent 29.7% of the samples to the donor group in a false-flag attack. PIFA keeps this descriptive signal without the vulnerability. Because it is trained on forged, copied, and edited copies of every executable, a split on a descriptive field gains nothing when the field is manipulated, so the trees fall back on the loader-relevant structure an adversary cannot change. It therefore stayed near 91.2% under forging, false flags, and edits, where the clean all-header trees collapsed and the loader-relevant trees fell under editing. Adversarial training thus reconciles the accuracy of the full header with robustness, at no inference cost.

A learned representation of the raw file did not match the header-field trees. On protected test executables, ObfusFormer was 23.4 points above the most accurate baseline network but below the header-field trees, and standardization instead of the bin encoding widened the gap. Gorishniy *et al.* [39] reported that piecewise-linear encodings of numerical features improved Transformers on tabular data, and Grinsztajn *et al.* [44] that tree ensembles remained more accurate than deep networks, including a Transformer, on medium-sized tabular data; the header fields are such tabular data.

6.2 Remaining Limitations

All protected executables came from one protector, VMProtect, which Han *et al.* [13] applied with options that their paper does not report. The archive of their study contains version 3.5.0. Other virtualizers such as Themida and Code Virtualizer, and protector options that pack the output file, may rebuild the section table and remove the layout that the loader-relevant fields describe. We forged header bytes but copied and edited fields only in the parsed headers, and we did not write or execute any manipulated file. The edits therefore assume that the changed flags and stack and heap sizes leave a program runnable. Larger functionality-preserving changes, such as injected or padded sections [30], [31], would alter the layout fields and were not evaluated. The split by time orders builds by their compile time stamps, which an author can set. The APT labels come from indicators in threat-intelligence reports, and Gray *et al.* [4] noted a lack of assurance about how such labels were assigned. The detection images render EMBER feature vectors rather than raw files, so the detection experiment did not test the structure view or the windows. These data also contain no protected benign software, so we could not measure whether protection alone triggers a malicious verdict. Protecting the training executables with several virtualizers and option sets would show whether loader-relevant header fields still identify a family when the protector rebuilds the header. Injecting and padding sections of the test executables would test attribution against layout edits.

7 Conclusion

We have proposed PIFA, a gradient-boosted attributor that reads the full portable executable header but is trained on forged, copied, and edited copies of every executable, so that it uses a field only where the field agrees with the loader-relevant structure that an adversary cannot change. On unseen builds of nine APT groups, PIFA has attributed VMProtect-protected executables with a macro-F_1 of 91.2% in seconds per fold, and has stayed at 91.1%, 90.9%, and 91.0% under forgery, false-flag, and editing attacks, the only method we tested that resists all three. Clean-trained trees on all header fields were more accurate on untouched samples (92.5%) but a false flag has cut them to 50.2%, and trees on the loader-relevant fields alone have stayed robust (90.9%) but fell to 85.3% under editing. PIFA has been more accurate than the multi-view Transformer we have developed (ObfusFormer, 87.5%) and every image and byte network, although its accuracy rests in part on related builds appearing in training and fell on later builds, which our leakage-free, build-grouped, adversary-aware benchmark measures.

In our future work, we will protect executables with several virtualizers and option sets and date builds by first-seen times, to test whether the loader-relevant header fields identify the families of new builds when the protector rebuilds the header.

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